

C957

Applied Algebra

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A nonprofit founded by two nurses tracks its members: 500 in year 0, 550 in year 1, 605 in year 2 and 665.5 in year 3. Memberships renew each January. Which model fits the data?

- A. Logistic, because membership must level off
- B. Linear, adding 55 members per year
- C. Linear, adding 50 members per year
- D. Exponential, growing 10% per year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which table of values (at $x = 0, 1, 2, 3$) represents a linear function?

- A. 40, 55, 70, 90
- B. 40, 55, 70, 85
- C. 40, 50, 65, 85
- D. 40, 60, 90, 135

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A laptop bought for \$1,200 for a sales rep in Denver loses 20% of its value each year. It came with a two-year warranty. What is its value after 3 years?

- A. \$2,073.60
- B. \$480.00
- C. \$614.40
- D. \$960.00

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An account grows exponentially from \$2,000 to \$2,420 in 2 years. What is the annual growth rate?

- A. 10.5%
- B. 110%
- C. 10%
- D. 21%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A shop that sells handmade backpacks at three craft fairs models monthly profit as $P(x) = -2x^2 + 240x - 3,000$, where x is the price in dollars. What price maximizes profit?

- A. -\$60
- B. \$60
- C. \$4,200
- D. \$120

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

For $P(x) = -2x^2 + 240x - 3,000$, the maximum occurs at $x = 60$. What is the maximum profit?

- A. -\$3,000
- B. \$60
- C. \$18,600
- D. \$4,200

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A three-person startup that raised a \$2 million seed round models app users (in thousands) as $f(t) = t^2 + 4t$, where t is months since launch. What is the average rate of change from month 2 to month 5?

- A. 11 thousand users per month
- B. 9 thousand users per month
- C. 14 thousand users per month
- D. 33 thousand users per month

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

After a product launch covered by a trade magazine, monthly sales were 100, 130, 150, 160 and 165 units. Which description fits?

- A. Increasing and concave up
- B. Linear, because it rises every month
- C. Decreasing, because growth is slowing
- D. Increasing and concave down

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In a town of 8,000 residents, the number who have adopted a new payment app is modeled by $P(t) = 5,000 / (1 + 49e^{(-0.3t)})$, where t is in months. In the long run, how many residents does the model predict will use the app?

- A. 5,000
- B. No limit; it keeps growing
- C. 8,000
- D. 100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

For $P(t) = 5,000 / (1 + 49e^{(-0.3t)})$, at what number of users is the growth rate fastest?

- A. Growth is always fastest at the start.
- B. 100
- C. 2,500
- D. 5,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A food truck that operates Friday through Sunday has revenue $R(x) = 25x$ and cost $C(x) = 8x + 3,400$, where x is meals sold per month. What is the profit when it sells 300 meals?

- A. \$5,100
- B. \$1,700
- C. -\$1,700
- D. \$7,500

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Coffee served in a ceramic mug at a café that opened last spring cools according to $T(t) = 70 + 110(0.9)^t$, in °F, where t is minutes. What temperature does the coffee approach over time?

- A. 70°F
- B. 0°F
- C. 180°F
- D. 110°F

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A nonprofit ~~founded by two nurses~~ tracks its members: 500 in year 0, 550 in year 1, 605 in year 2 and 665.5 in year 3. ~~Memberships renew each January.~~ Which model fits the data?

Target word: model type **Decides:** Each year is 1.10 times the last; differences grow.

A	Adjacent category	Nothing in the data shows leveling off.
B	Wrong stem number	Uses one middle difference; differences are not constant.
C	Dropped element	Checks only the first difference; later differences are 55 and 60.5.
D	Correct	$550 / 500 = 605 / 550 = 665.5 / 605 = 1.1$.

2. Answer B Concept 1 • transfer

Which table of values (at $x = 0, 1, 2, 3$) represents a linear function?

Target word: identify linear from the rule, reversed **Decides:** Constant difference.

A	Dropped element	Checking only the first two differences misses the final jump of 20.
B	Correct	Adds 15 every step.
C	Adjacent category	Differences 10, 15, 20 grow steadily: quadratic.
D	Confusion set	Multiplies by 1.5 each step: exponential.

3. Answer C Concept 2

A laptop bought for \$1,200 ~~for a sales rep in Denver~~ loses 20% of its value each year. ~~It came with a two-year warranty.~~ What is its value after 3 years?

Target word: value after 3 years **Decides:** $1,200 \times 0.8^3$.

A	Inverted operation	Uses a growth factor of 1.2 instead of 0.8.
B	Adjacent category	Subtracts \$240 each year: linear, not percentage, depreciation.
C	Correct	$0.8^3 = 0.512$; $1,200 \times 0.512 = 614.40$.
D	Dropped element	Applies the decay for only one year.

4. Answer C Concept 2 • transfer

An account grows exponentially from \$2,000 to \$2,420 in 2 years. What is the annual growth rate?

Target word: solve for r **Decides:** $(1 + r)^2 = 2,420 / 2,000 = 1.21$.

A	Adjacent category	$420 / 2 / 2,000$ assumes linear growth.
B	Confusion set	Reads the growth factor 1.1 as the rate.
C	Correct	$\sqrt{1.21} = 1.1$, so $r = 0.10$.
D	Dropped element	The total two-year change, not annual.

5. Answer B Concept 3

A shop ~~that sells handmade backpacks at three craft fairs~~ models monthly profit as $P(x) = -2x^2 + 240x - 3,000$, where x is the price in dollars. What price maximizes profit?

Target word: input at the vertex **Decides:** $x = -240 / (2 \times -2)$.

A	Inverted operation	Sign error on $-b$.
B	Correct	$-240 / -4 = 60$.
C	Adjacent category	The maximum profit, not the price.
D	Dropped element	$-b / a$ drops the 2 in $2a$.

6. Answer D Concept 3 • transfer

For $P(x) = -2x^2 + 240x - 3,000$, the maximum occurs at $x = 60$. What is the maximum profit?

Target word: solve for the output at the vertex **Decides:** $P(60)$.

A	Wrong stem number	$P(0)$, the constant term.
B	Adjacent category	The price, not the profit.
C	Inverted operation	Adds $2x^2$ instead of subtracting it.
D	Correct	$-2(3,600) + 14,400 - 3,000 = 4,200$.

7. Answer A Concept 4

A ~~three-person startup that raised a \$2-million seed round~~ models app users (in thousands) as $f(t) = t^2 + 4t$, where t is months since launch. What is the average rate of change from month 2 to month 5?

Target word: average rate on $[2, 5]$ **Decides:** $(f(5) - f(2)) / 3$.

A	Correct	$f(5) = 45$, $f(2) = 12$; $33 / 3 = 11$.
B	Wrong stem number	$f(5) / 5$ ignores the starting point.
C	Adjacent category	The instantaneous rate at month 5, not the average.
D	Dropped element	Forgets to divide by 3 months.

8. Answer D Concept 5

After a product launch ~~covered by a trade magazine~~, monthly sales were 100, 130, 150, 160 and 165 units. Which description fits?

Target word: direction and concavity **Decides:** Gains 30, 20, 10, 5 shrink while sales rise.

A	Inverted operation	Concave up needs growing increases.
B	Absolute language	Rising every month does not make it linear; gains are not constant.
C	Confusion set	Confuses a falling rate with falling sales.
D	Correct	Rising, but by smaller amounts each month.

9. Answer A Concept 6

In a town of ~~8,000 residents~~, the number who have adopted a new payment app is modeled by $P(t) = 5,000 / (1 + 49e^{(-0.3t)})$, where t is in months. In the long run, how many residents does the model predict will use the app?

Target word: carrying capacity **Decides:** As t grows, the denominator approaches 1.

A	Correct	The numerator is the carrying capacity.
B	Adjacent category	That describes exponential, not logistic, growth.
C	Camouflage	The town size is not in the model.
D	Confusion set	$P(0) = 5,000 / 50$, the starting value.

10. Answer C Concept 6 · transfer

For $P(t) = 5,000 / (1 + 49e^{(-0.3t)})$, at what number of users is the growth rate fastest?

Target word: inflection level **Decides:** Logistic growth is fastest at half the carrying capacity.

A	Absolute language	Logistic growth starts slow, speeds up, then slows; it peaks at half capacity.
B	Wrong stem number	The starting value.
C	Correct	Half of 5,000.
D	Inverted operation	Growth nearly stops near capacity.

11. Answer B Concept 7

A food truck ~~that operates Friday through Sunday~~ has revenue $R(x) = 25x$ and cost $C(x) = 8x + 3,400$, where x is meals sold per month. What is the profit when it sells 300 meals?

Target word: $P(300)$ **Decides:** $P(x) = 17x - 3,400$.

A	Adjacent category	17×300 is the contribution before fixed cost.
B	Correct	$7,500 - (2,400 + 3,400) = 1,700$.
C	Inverted operation	$C - R$ reverses the subtraction.
D	Dropped element	Revenue only; leaves out cost.

12. Answer A Concept 8

Coffee served ~~in a ceramic mug at a café that opened last spring~~ cools according to $T(t) = 70 + 110(0.9)^t$, in °F, where t is minutes. What temperature does the coffee approach over time?

Target word: horizontal asymptote **Decides:** $110(0.9)^t$ goes to 0, leaving 70.

A	Correct	The decaying term vanishes; T approaches the added constant.
B	Dropped element	Drops the added 70; not every decay heads to zero.
C	Wrong stem number	$T(0)$, the starting temperature.
D	Confusion set	The starting gap above room temperature, not a limit.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Linear vs. exponential patterns	Linear change adds the same amount each step (constant difference); exponential change multiplies by the same factor (constant ratio).	Checking only the first difference, or calling any curve-like growth exponential.	A table of values at equal steps.
2	Exponential growth and decay	$y = a(1 + r)^t$; growth factor $1 + r$ above 1, decay factor $1 - r$ below 1.	Subtracting a fixed amount each year, or using $1 + r$ for decay.	"Grows by x% per year" or "loses x% per year".
3	Quadratic maximum or minimum	For $ax^2 + bx + c$ the vertex is at $x = -b / (2a)$; $a < 0$ gives a maximum.	Dropping the 2, flipping the sign, or reporting the output when the input is asked.	A profit or area function with an x^2 term.
4	Average rate of change	Average rate of change on $[a, b] = (f(b) - f(a)) / (b - a)$.	Forgetting to divide by the interval length, or using an instantaneous rate.	"Average rate from $t = a$ to $t = b$ ".
5	Concavity	Concave up = rate of change increasing; concave down = rate of change decreasing, whether the function rises or falls.	Reading shrinking gains as a falling function.	Increases that get smaller or larger each step.
6	Logistic growth	Logistic growth levels off at the carrying capacity (the numerator of $L / (1 + Ae^{(-kt)})$) and is fastest at half that level.	Treating it as unlimited exponential growth, or assuming growth is fastest at the start.	An S-shaped model or a function of the form $L / (1 + Ae^{(-kt)})$.
7	Combining functions	Profit $P(x) = R(x) - C(x)$; subtract the whole cost function, fixed and variable parts.	Reversing the subtraction or leaving out fixed cost.	Separate revenue and cost functions.
8	Horizontal asymptotes of shifted exponentials	For $y = c + a \cdot b^t$ with $0 < b < 1$, y approaches c as t grows.	Assuming every decay heads to zero.	A constant added to a decaying exponential term.

Course scope (public description): Applied Algebra is designed to help you develop competence in working with functions, the algebra of functions, and using some applied properties of functions. You will start learning about how we can apply different kinds of functions to relevant, real-life examples. From there, the algebra of several families of functions will be explored, including linear, polynomial, exponential, and logistic functions. You will also learn about relevant, applicable mathematical properties of each family of functions, including rate of change, concavity, maximizing/minimizing, and asymptotes. These properties will be used to solve problems related to your major and make sense of everyday living problems. Students should complete Applied Probability and Statistics or its equivalent prior to engaging in Applied Algebra.